Supporting Subjective Image Queries without Seeding Requirements — Proposing Test Queries for Benchathlon

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ABSTRACT

Traditional content-based image retrieval (CBIR) systems suffer from at least two shortcomings. First, most systems require users to provide "good" images to initialize queries. We argue that finding good seeds is the job of the search engine itself, and this circular requirement leaves the core problem—understanding users' query concepts—unsolved. The second shortcoming is that most systems fail to adequately model users' subjectivity. This paper proposes two test queries to Benchathlon for steering the CBIR community to seriously address the above issues. We also present a search scenario to show how such queries may be supported and their performance be measured.

Keywords: content-based, perception-based, image retrieval, benchmark

1. INTRODUCTION

BIRDS-1 marks a step forward for establishing a standardized benchmark for measuring the performance of content-based image retrieval (CBIR) systems.[1,2,?,3–8] However, the sample queries proposed in BIRDS-1 may not adequately reflect how users typically search for visual content. First, query types #1 and #4 may require users to provide seed images to start a query. We argue that finding good seeds is the job of the search engine itself, and this circular requirement leaves the core problem—understanding users' query concepts—unsolved. Second, the benchmark does not seem to consider subjective perception,[9–11] that is, different users may perceive the same image differently and may use different vocabulary to describe it. It may not be realistic to expect that there is only one ground truth for measuring image "similarity."

To address the above shortcomings, we have proposed a new image search paradigm: perception-based image retrieval (PBIR),[12] which requires no seeding and which measures similarity considering individuals' subjectivity. A PBIR query iterates the following three steps until the user terminates the query:

1. The system presents a set of image samples to the user. The user marks the images that are relevant to his/her query concept.
2. The system learns the query concept based on the feedback, and report the top-k most similar images based on the learned concept at this point. Continue to Step 3 unless the query is terminated.
3. The system further refines the bound to perform the next round of sample generation. The samples selected in the refined sampling space are the ones that can collect maximum information from the user to learn the target concept. Return to Step one.

To measure the performance of a PBIR query, we propose the following two test queries.

- Query #1. How fast to find a target image R. Insert image R into the database. Without seeding, use the search method provided by the system to find R in the database.
- Query #2. Find images that satisfy a query concept C. A concept may be a category like landscape, architecture, or flowers, and can be an abstract concept like dancing. The system returns images that are related to the concept.

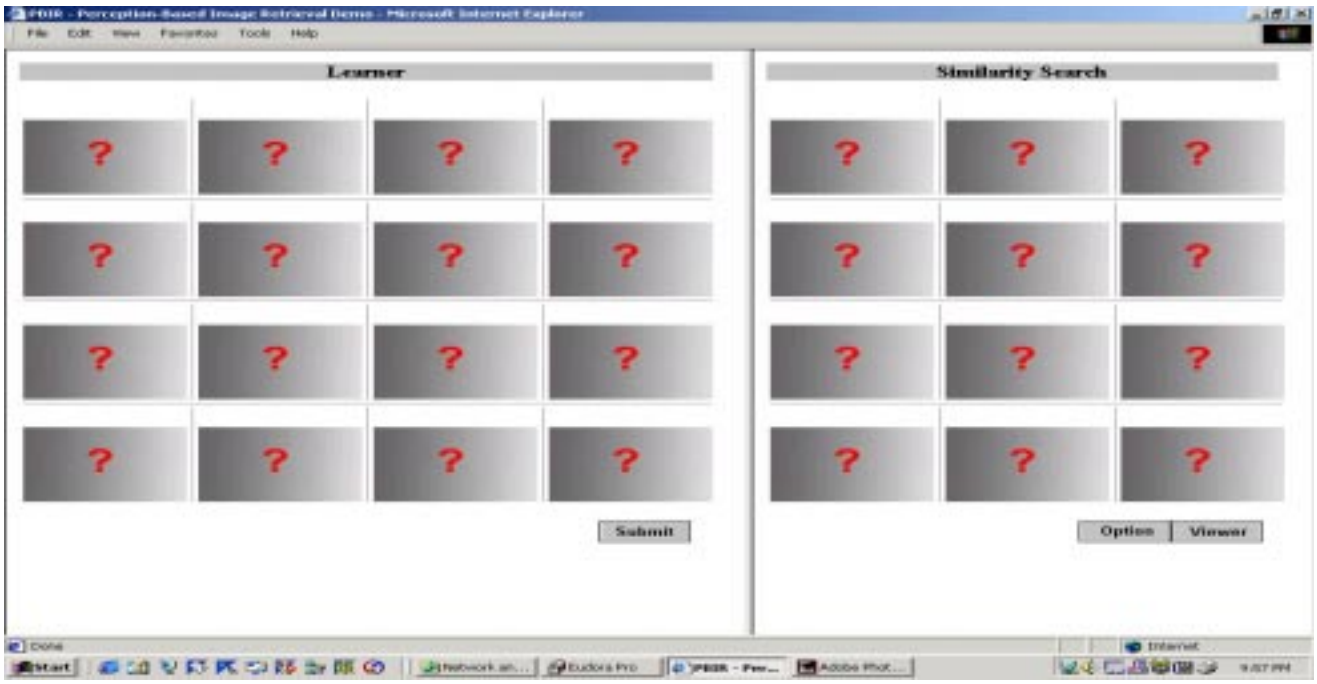


Figure 1. Wild Animal Query Screen #1.

In brief, it is critical for a multimedia search engine to take individuals' subjectivity into consideration. It is important not to expect "perfect samples" can be provided by users to start a query. Therefore, the standard benchmark should include test queries like the ones proposed above to measure a system's capability to fulfill these two requirements.

The rest of this paper is organized into four sections. Section 2 walks through an example query of our PBIR system to illustrate how the above queries may be supported. The testbed can provide a similar interface to test a CBIR system. Section 3 specifies the queries we propose and the metrics for measuring their performance. Finally, we offer our conclusion remarks in Section 4.

2. ILLUSTRATIVE EXAMPLES

We present an example query in this section to show the learning steps of MEGA[13–15] and SVM_{Active}[16] in two image query scenarios: *image browsing* and *similarity search*.

- *Image browsing*. A user knows what he/she wants but has difficulty articulating it. Through an interactive browsing session, our active learning algorithms, MEGA and SVM_{Active}, learn what the user wants.
- *Similarity search*. After MEGA and SVM_{Active} capture what the user wants, the search engine can perform a traditional similarity search to find data objects that appear similar to a given query object.

2.1. Query Steps

This interactive query session involves seven screens that are illustrated in seven figures. The user's query concept in this example is "wild animals."

1. Screen 1. Initial Screen. Our Perception-Based Image Retrieval (PBIR) system presents the initial screen to the user as depicted in Figure 1. The screen is split into two frames vertically. On the left-hand side of the screen is the learner frame; on the right-hand side is the similarity search frame. Through the learner frame, PBIR learns what the user wants via an intelligent sampling process. The similarity search frame

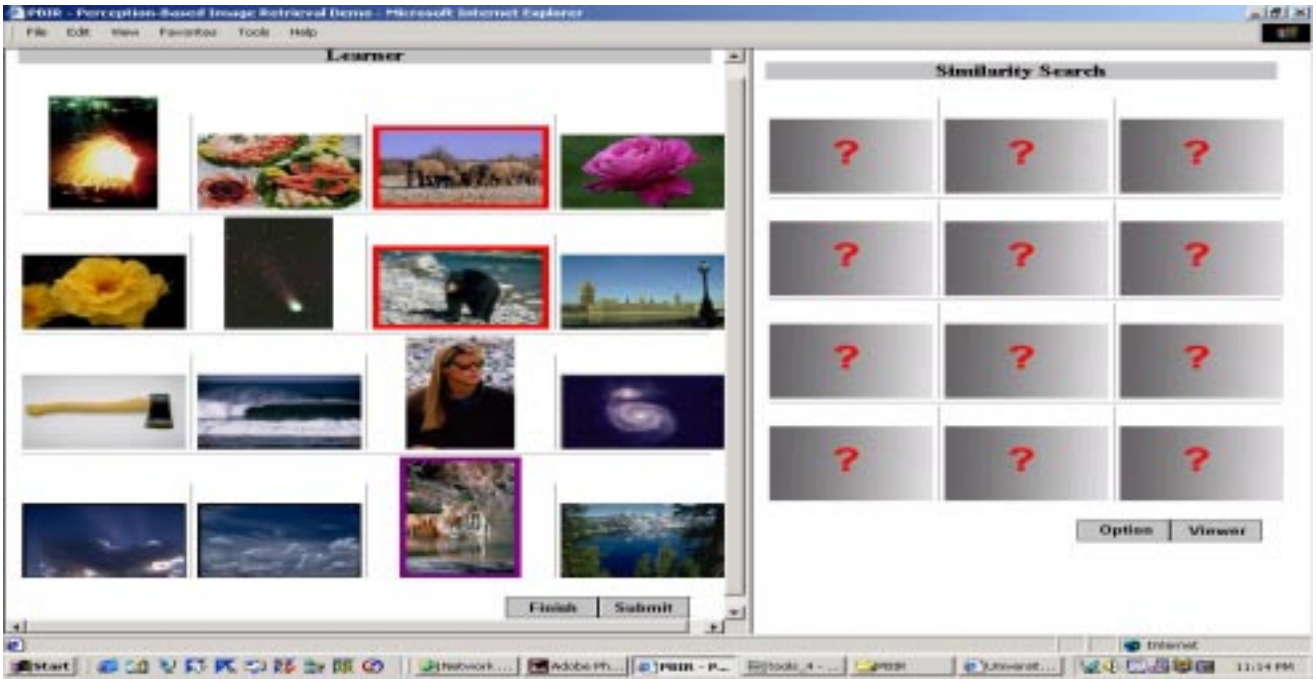


Figure 2. Wild Animal Query Screen #2.

displays what the system thinks the user wants. (The user can set the number of images to be displayed in these frames.)

2. Screen 2. Sampling and relevance feedback starts. Once the user clicks the "submit" button in the initial frame, the sampling and relevance feedback step commences to learn what the user wants. The PBIR system presents a number of samples in the learner frame, and the user highlights images that are relevant to his/her query concept by clicking on the relevant images. As shown in Figure 2, three images (the third image in rows one, two and four in the learner frame) are selected as relevant, and the rest of the unmarked images are considered irrelevant. The user indicates the end of his/her selection by clicking on the submit button in the learner screen. This action brings up the next screen.

3. Screen 3. Sampling and relevance feedback continues. Figure 3 shows the third screen. At this time, the similarity search frame still does not show any image, since the system has not been able to grasp the user's query concept at this point. The PBIR system again presents samples in the learner frame to solicit feedback. The user selects the second image in the third row as the only image relevant to the query concept.

4. Screen 4. Sampling and relevance feedback continues. Figure 4 shows the fourth screen. First, the similarity search frame displays what the PBIR system thinks will match the user's query concept at this time. As the figure indicates, the top nine returned images fit the concept of "wild animals." The user's query concept has been captured, though somewhat fuzzily. The user can ask the system to further refine the target concept by selecting relevant images in the learner frame. In this example, the fourth image in the second row and the third image in the fourth row are selected as relevant to the concept. After the user clicks on the submit button in the learner frame, the fifth screen is displayed.

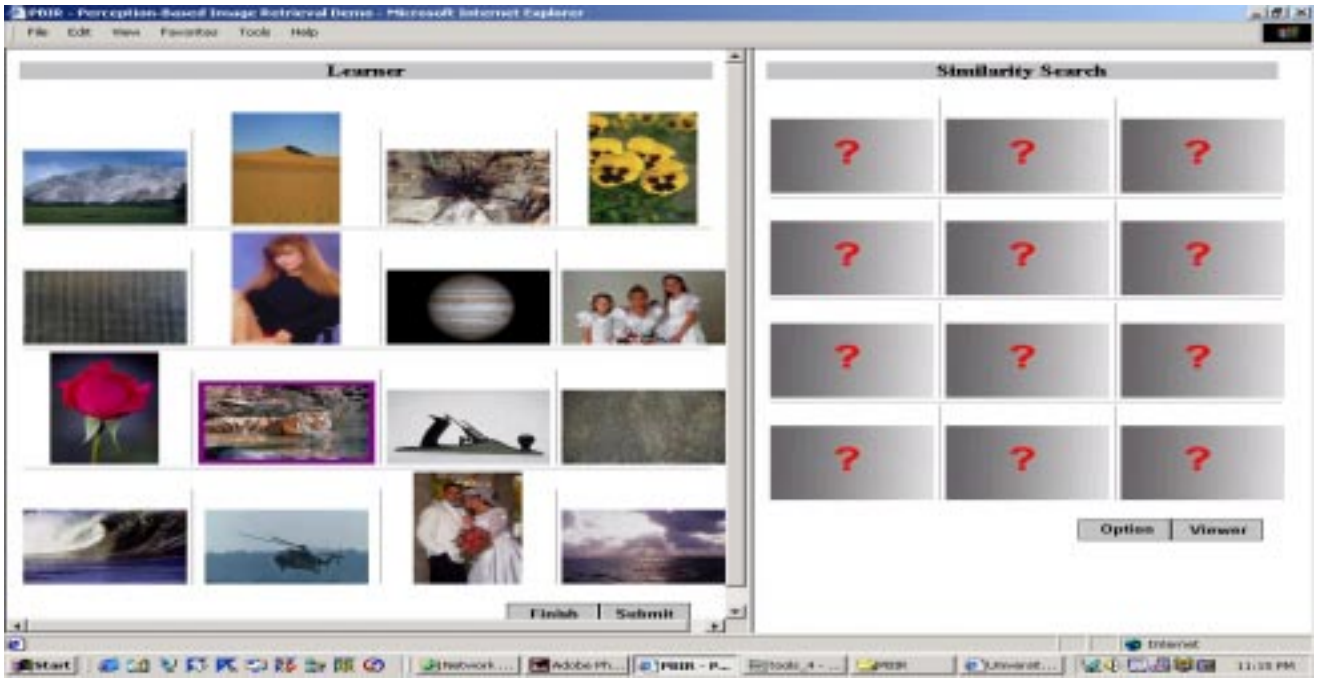


Figure 3. Wild Animal Query Screen #3.

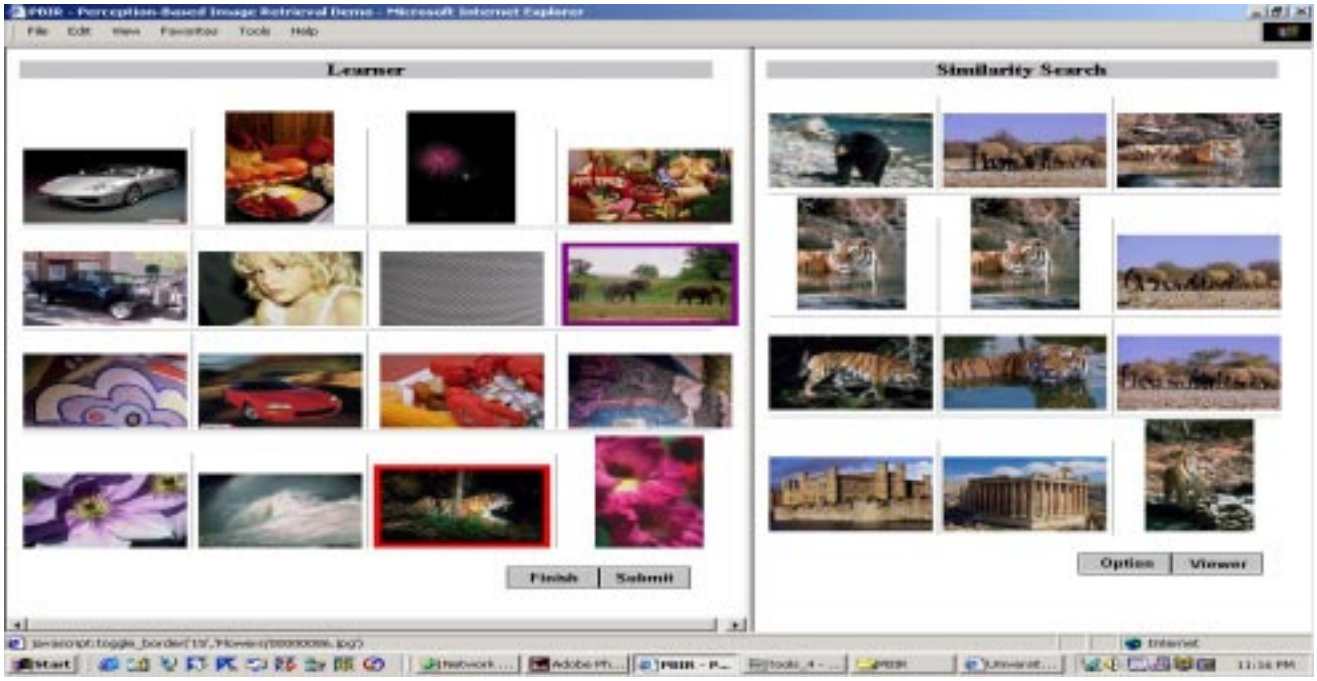


Figure 4. Wild Animal Query Screen #4.



Figure 5. Wild Animal Query Screen #5.

5. Screen 5. Sampling and relevance feedback continues. The similarity search frame in Figure 5 shows that ten out of the top twelve images returned match the "wild animals" concept. The user selects four relevant images displayed in the learner frame. This leads to the final screen of this learning series.

6. Screen 6. Sampling and relevance feedback ends. Figure 6 shows that all returned images in the similarity search frame fit the query concept (wild animals).

7. Screen 7. Similarity search. At any time, the user can click on an image in the similarity search frame to request images that *appear similar* to the selected image. This step allows the user to zoom in onto a specific set of images that match some appearance criteria, such as color distribution, textures and shapes. As shown in Figure 7, after clicking on one of the tiger images, the user will find similar tiger images returned in the similarity search frame. Notice that other wild animals are ranked lower than the matching tiger images, since the user has concentrated more on specific appearances than on general concepts.

This example demonstrates a potential to facilitate information retrieval in two respects:

- *Eliminating the seeding requirement.* One is not required to initiate a query with "good" seed images. Suppose a victim wants to find the perpetrators, he/she does not need to provide an example image to start the search. Even if all initial examples presented are negative, the negative-labeled instances can be used to shrink the pool of the next sampling round substantially. Therefore, the probability that a relevant example will be presented in the next iteration will be substantially higher.

- *Supporting flexible concept formulation.* One can search for a general animal concept or a specific kind of animal (e.g., tigers). In the above example, if we were to mark only tiger images as relevant, the final result would show tiger images only. Also note in the third screen that the system presents animals of different



Figure 6. Wild Animal Query Screen #6.



Figure 7. Wild Animal Similarity Query (Screen #7).

kinds, alone or in groups, and on different backgrounds to ask for feedback. The system can intelligently explore more possibilities to refine the query concept.

3. TEST QUERIES

3.1. Query #1: How Fast to Find Target Image R

Insert image R into the database. Use the search method(s) (e.g., query languages, keywords, and drawlings) provided by the system to find R in the database. This query should be performed without any example images.

- Goal: Finding a specific image.
- Input: Image R.
- Output: The number of images browsed, N, before finding R in the dataset.
- Steps and metrics:

Step	*Description*	*Metric*
#1	*Insert image R into the database*	*Time*
#2	*Find C*	*N, the number of images browsed, Time*

3.2. Query #2: Find Images Satisfying Query Concept C

Returns k images that are related to the concept C. A concept can be a broad image category such as landscape or animals, and can be a specific concept such as fighter jets. This query does not accept any explicit input. A user refines his/her search through relevance feedback.

- Goal: Finding images matching query concept C.
- Input: Number of relevance feedback rounds, M; Number of returned results k.
- Output: Image set C'.
- Steps and metrics:

Step	*Description*	*Metric*
#1	*Relevance feedback starts*	
$\cdots$	*Relevance feedback continues*	*Precision, Time*
M	*Relevance feedback ends*	*Precision, Time*

4. CONCLUSION

It is not realistic to expect "perfect samples" can be provided by users to start an image query. It is critical for a multimedia search engine to take individuals' subjectivity into consideration. Therefore, the standard benchmark should include test queries that measure a system's capability to fulfill these two requirements. In this paper we proposed two such test queries and the metrics to measure their performance. We also presented an example to demonstrate how such queries can be supported.

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